

Mahila-Shakti Unnati

Micro-Finance Digital Ledger for Women's Self-Help Groups
Android App Development Using GenAI | Empowering Rural Women Through Technology

5+	3	100%	∞
Core Modules	User Roles	Offline-First	Impact Potential

1. Problem Statement

Self-Help Groups (SHGs) are the backbone of rural micro-finance in India, empowering millions of women to pool savings and access credit. However, nearly all SHGs still rely on **physical registers and manual bookkeeping** — a process riddled with risk.

- **Error-Prone Ledgers:** Manual arithmetic mistakes in weekly savings and interest calculations lead to incorrect balances.
- **Dispute Triggers:** Discrepancies in handwritten records create mistrust and community conflict.
- **No Audit Trail:** There is no reliable way to verify historical transactions.
- **Loan Mismanagement:** Double loans and unpaid dues go undetected without a centralized system.
- **Zero Digital Footprint:** Members cannot build credit histories for formal banking access.

2. Vision & Detailed Description

Mahila-Shakti Unnati is a fully offline-capable Android application that serves as a *'Digital Accountant'* for women's SHGs. Built with Kotlin and Room DB, it replaces paper registers with a tamper-resistant digital ledger accessible even without internet.

The app is designed to be operated by semi-literate rural women using a simple, icon-driven interface in regional languages. It automatically calculates savings totals, simple interest, and loan eligibility — removing human error entirely.

3. Scope of the Application

IN SCOPE	OUT OF SCOPE
Member registration with photo & details	Real-time bank integration or UPI payments
Weekly savings entry (Paid / Pending)	Multi-group federated management dashboard
Loan issuance, tracking & interest computation	Cloud sync or remote server backend

WhatsApp / text export of financial summaries	Biometric authentication
Group capital and loan eligibility calculation	Regulatory / RBI compliance reporting
Offline-first local Room DB storage	Automated SMS or notification to members
Admin + Member role differentiation	Formal credit score bureau reporting

4. Functional Requirements

ID	Feature	Requirement Description
FR-01	Member Management	The system shall allow an admin to add, edit, and deactivate SHG members with name, photo, phone, and joining date.
FR-02	Savings Entry	The system shall record each member's weekly savings as 'Paid' or 'Pending' with timestamp.
FR-03	Savings Aggregation	The system shall automatically recalculate total group savings and individual balances upon each new entry.
FR-04	Loan Issuance	The system shall allow a loan to be issued only if the member has no existing unpaid loan.
FR-05	Interest Calculation	The system shall compute simple interest on active loans using configurable rate and duration.
FR-06	Repayment Tracking	The system shall record partial and full loan repayments and update outstanding balance in real time.
FR-07	Loan Eligibility Check	The system shall calculate and display each member's loan eligibility based on their savings contribution ratio.
FR-08	Data Export	The system shall export a formatted text summary of group financials for sharing via WhatsApp Intent.
FR-09	Contribution History	The system shall display a chronological payment history for each member.
FR-10	Role Access	Admin users shall have full access; member-view users shall see only their own data.

5. Non-Functional Requirements

Category	Requirement	Target / Metric
Performance	Savings total must update on new entry	< 500 ms response time
Offline Capability	Full functionality without internet	100% offline (Room DB)
Reliability	No data loss on app crash or restart	ACID transactions via Room
Usability	Operable by semi-literate users	Icon-driven UI, regional lang. support

Scalability	Handle up to 50 members per group	No performance degradation
Security	PIN-protected admin access	Local encrypted storage
Maintainability	DB schema changes must not corrupt data	Room migration scripts required
Portability	Run on low-end Android devices	Min SDK: Android 6.0 (API 23)

6. Feature Catalogue

■ MUST-HAVE FEATURES	
Feature	Description
Member Directory	Add members with name, photo, phone, and role. Edit or soft-delete members without losing history.
Weekly Savings Entry	Mark each member's weekly contribution as Paid or Pending. Auto-timestamp every record.
Real-Time Savings Total	Group total savings and individual balances recalculate instantly on every new savings entry.
Loan Issuance with Guard	Issue loans to members. System blocks new loan if an unpaid loan already exists for that member.
Simple Interest Engine	Compute simple interest ($P \times R \times T / 100$) on active loans. Configurable interest rate per group.
Repayment Tracker	Record partial/full repayments. Show outstanding principal + accrued interest at any time.
Loan Eligibility Display	Show each member's eligible loan amount based on their savings ratio vs. group capital.
WhatsApp Export	Generate a plain-text financial summary and share via Android Intent (WhatsApp, email, SMS).
Room DB with Migrations	All data stored in Room DB with proper foreign-key relations. Migration scripts prevent data loss on schema updates.
Contribution History View	Chronological list of all savings payments per member with date and status.

■ GOOD-TO-HAVE FEATURES	
Feature	Description
GenAI Financial Advisor	Integrate a lightweight on-device LLM or Gemini API call to answer members' finance questions in local language.
PIN / Fingerprint Lock	Protect admin access with a PIN or Android BiometricPrompt for data security.
Multilingual UI	Support Hindi, Tamil, Telugu, Kannada using Android string resources and locale switching.

PDF Report Generation	Export monthly statements as a formatted PDF using iTextG or similar library.
Dark Mode	Full Material You dark-theme support for low-light rural environments.
Notification Reminders	Weekly WorkManager reminders to the admin to collect and enter savings dues.
Analytics Dashboard	Bar/pie charts showing savings trends, loan utilization, and repayment rates using MPAndroidChart.
Backup to Google Drive	Optional encrypted JSON backup to the group's shared Google Drive folder.
Credit Score Indicator	Simple in-app credit score (0–100) derived from repayment consistency to teach financial literacy.
Aadhaar / Member ID Linking	Optional field to store masked Aadhaar or member ID for identity verification.

7. App Usage & User Flow

Step	Action	Details
1	Admin Login	Admin unlocks the app with PIN. Dashboard shows group capital, pending dues, and active loans at a glance.
2	Member Directory	Navigate to Members tab. Tap '+' to add a new member — fill name, photo (camera/gallery), phone, and join date.
3	Weekly Savings Round	Go to Savings. Select the week. For each member, tap Paid or Pending. Totals update instantly.
4	Loan Application	Member requests loan. Admin opens Loans > New Loan, selects member — system checks for unpaid loans and shows eligibility. Enter amount and rate.
5	Repayment Entry	On repayment day, Admin selects active loan, enters amount received. Interest auto-accrues. Balance updates.
6	Export & Share	Tap Export on the Dashboard. A clean summary (members, savings, loans, balances) is formatted and shared via WhatsApp Intent.

8. Technical Implementation

Language:	Kotlin — primary language for all business logic, coroutines for async DB operations.
Architecture:	MVVM (Model-View-ViewModel) with LiveData / StateFlow for reactive UI updates.
Database:	Room DB — entities: Member, SavingsEntry, Loan, Repayment. One-to-Many relations (Member → SavingsEntries, Member → Loans).
Interest Logic:	Simple Interest = (Principal × Rate × Time) / 100, computed in Kotlin inside a ViewModel.

Export:	Android Implicit Intent (ACTION_SEND) with MIME type text/plain to share summary via WhatsApp.
GenAI Integration:	Optional: Gemini API or on-device ML Kit for natural language Q&A; about group finances.
DB Migrations:	Room @Migration classes handle schema version upgrades without data loss.
UI:	Material Design 3 components, RecyclerView with DiffUtil, CameraX for photo capture.

9. Impact Goals

Impact Area	How the App Delivers It
Women's Empowerment	Giving rural women digital tools to independently manage their collective capital, reducing dependency on male intermediaries.
Financial Literacy	The app's real-time interest display and credit-score indicator teach members the basics of credit and money management.
Transparency	An immutable digital ledger eliminates the ambiguity of handwritten registers, building trust within the group.
Scalability of Impact	Each SHG using the app can potentially link to formal banking, scaling micro-finance reach to thousands of women.

10. Success Criteria for Students

ID	Success Criterion
SC-01	Savings totals must update instantly (< 500 ms) when a new savings entry is added — no manual refresh.
SC-02	The app must display an error and block loan creation if the selected member has an existing unpaid loan.
SC-03	Data must be exportable as a clean, human-readable text string shareable via WhatsApp without extra formatting steps.
SC-04	Room DB migrations must be correctly implemented — upgrading the app must not crash or corrupt existing data.
SC-05	Simple interest calculation must be accurate to 2 decimal places for all standard inputs.
SC-06	Member photos must persist across app restarts and be displayed in the member list without reloading.

This document is a comprehensive project specification for the Mahila-Shakti Unnati Android application. Students are expected to implement all Must-Have features as part of core deliverables. Good-to-Have features may be considered for bonus credit or extended project scope.